



Yakeen Plus (2025)

Short Practice Test - 08

DURATION : 60 Minutes

DATE : 10/11/2024

M. MARKS : 192

ANSWER KEY

PHYSICS

1. (1)
2. (2)
3. (3)
4. (4)
5. (2)
6. (1)
7. (3)
8. (4)
9. (2)
10. (3)
11. (4)
12. (1)

CHEMISTRY

13. (2)
14. (3)
15. (2)
16. (3)
17. (1)
18. (4)
19. (2)
20. (4)
21. (1)
22. (2)
23. (2)
24. (3)

BOTANY

25. (3)
26. (1)
27. (2)
28. (3)
29. (1)
30. (4)
31. (3)
32. (2)
33. (1)
34. (2)
35. (1)
36. (3)

ZOOLOGY

37. (2)
38. (2)
39. (1)
40. (2)
41. (4)
42. (4)
43. (1)
44. (4)
45. (4)
46. (1)
47. (2)
48. (2)

SECTION-(I) PHYSICS

1. (1)

$$\frac{E_0}{B_0} = c \text{ also } k = \frac{2\pi}{\lambda} \text{ and } \omega = 2\pi\nu$$

These relation give $E_0 k = B_0 \omega$ [New NCERT Class 12th Page No. 272]

2. (2)

We know that quality factor (Q),

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}}$$

$$Q^2 = \frac{L}{R^2 C} \text{ or } C = \frac{L}{Q^2 R^2}$$

[New NCERT Class 12th Page No. 245]

3. (3)

Here, inductor also have internal resistance (r_L)So, total resistance (R) = 40 + 40 = 80 Ω The impedance of $L - C - R$ circuit is given by

$$Z = \sqrt{R^2 + (X_L - X_C)^2}$$

$$= \sqrt{(80)^2 + (100 - 40)^2} = 100 \Omega$$

Power factor = $R/Z = 80/100 = 0.8$ [New NCERT Class 12th Page No. 245]

4. (4)

$$n_1 \lambda_1 = n_2 \lambda_2 \Rightarrow 62 \times 5893 = n_2 \times 4358 \Rightarrow n_2 = 84$$

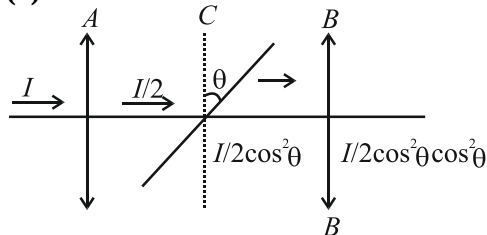
5. (2)

$$\text{Fringe width } W = \frac{2D\lambda}{a}$$

$$W \propto \frac{1}{a}$$

It means a increase and W decrease.

6. (1)

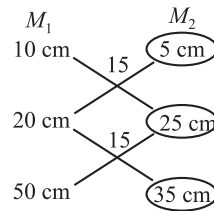
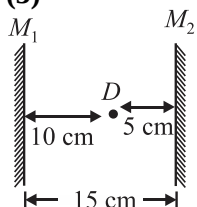


$$\frac{I}{2} \cos^4 \theta = \frac{I}{8}$$

$$\cos^4 \theta = \frac{1}{4}$$

$$\cos \theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ$$

7. (3)

[New NCERT Class 12th Page No. 222]

8. (4)

$$f = \frac{d^2 - x^2}{4d}$$

$$d = 80 + 40 = 120$$

$$x = 80 - 40 = 40$$

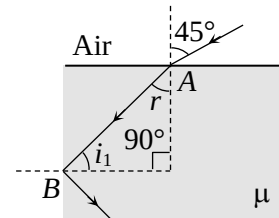
$$f = \frac{120^2 - 40^2}{4 \times 120} = \frac{80}{3} \text{ cm}$$

[New NCERT Class 12th Page No. 234]

9. (2)

At point A, by Snell's law

$$\mu = \frac{\sin 45^\circ}{\sin r} \Rightarrow \sin r = \frac{1}{\mu\sqrt{2}} \quad \dots(i)$$

At point B, for total internal reflection $\sin i_1 = \frac{1}{\mu}$ From figure, $i_1 = 90^\circ - r$ 

$$\therefore \sin(90^\circ - r) = \frac{1}{\mu}$$

$$\Rightarrow \cos r = \frac{1}{\mu} \quad \dots(ii)$$

$$\text{Now } \cos r = \sqrt{1 - \sin^2 r} = \sqrt{1 - \frac{1}{2\mu^2}}$$

$$= \sqrt{\frac{2\mu^2 - 1}{2\mu^2}} \quad \dots(iii)$$

From equation (ii) and (iii)

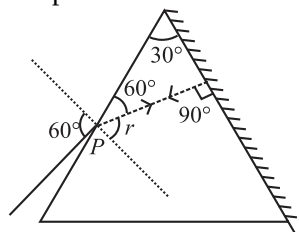
$$\frac{1}{\mu} = \sqrt{\frac{2\mu^2 - 1}{2\mu^2}}$$

Squaring both side and then solving we get

$$\mu = \sqrt{\frac{3}{2}}$$

[New NCERT Class 12th Page No. 239]

10. (3)
An isosceles prism



For retreat of path light should fall perpendicularly on silvered surface

For refraction at point P,

$$1 \sin 60^\circ = \mu \sin r$$

$$\mu \sin r = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \mu \sin 30^\circ = \frac{\sqrt{3}}{2}$$

$$\Rightarrow \frac{\mu}{2} = \frac{\sqrt{3}}{2} \Rightarrow \mu = \sqrt{3}$$

[New NCERT Class 12th Page No. 239]

11. (4)

$$\frac{E_0}{B_0} = c, B_0 = \frac{6 \times 10^6}{3 \times 10^8} = 2 \times 10^{-2} \text{ T}$$

[New NCERT Class 12th Page No. 276]

12. (1)

According to modified Ampere's circuital law.

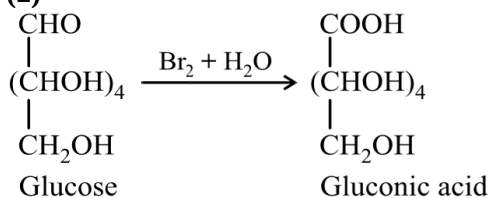
$$\oint \vec{B} \cdot d\vec{l} = \mu_0 \left(i_C + \epsilon_0 \frac{d\phi_E}{dt} \right)$$

[New NCERT Class 12th Page No. 205]

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SECTION-(II) CHEMISTRY

13. (2)



[New NCERT Class 12th Page No. 283]

14. (3)

Uracil is the only base present in RNA but not in DNA. Thymine, adenine, and guanine are found in both DNA and RNA.

[New NCERT Class 12th Page No. 298]

15. (2)

Sugars in which aldehyde or ketone functional groups are not free are called non-reducing sugars. In the case of sucrose, the free aldehyde is not present.

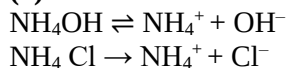
[New NCERT Class 12th Page No. 282]

16. (3)

Ceric ammonium nitrate test does not distinguish among all the three types of alcohols.

[Newly added topic]

17. (1)



Due to common ion (NH_4^+) effect, the above equilibrium is shifted to backward direction i.e. the dissociation of NH_4OH is suppressed. These precipitation of higher group cations is prevented.

[Newly added topic]

18. (4)

Double salts are combinations of two individual salts crystallized together in a specific ratio. They maintain their ionic identities and dissociate completely in water.

Complex salts are chemical compounds where a central metal ion forms coordination bonds with surrounding ligands (ions or molecules). They have a distinct structure and often exhibit unique properties different from their constituent components.

$4\text{KCN} \cdot \text{Fe}(\text{CN})_2$ is an example of complex salt.

[New NCERT Class 12th Page No. 119]

19. (2)

Both are outer orbital complexes and have octahedral geometry and sp^3d^2 state of hybridization

[New NCERT Class 12th Page No. 129]

20. (4)

IUPAC name of the complex compound $[\text{Co}(\text{NH}_3)_4(\text{H}_2\text{O})\text{Br}](\text{NO}_3)_2$ is Tetraammineaquabromidocobalt(III) nitrate.

[New NCERT Class 12th Page No. 123]

21. (1)

K_2MnO_4 has +6 oxidation state of Mn.

[New NCERT Class 12th Page No. 96]

22. (2)

+3 oxidation state is most common among all the Lanthanoids.

[New NCERT Class 12th Page No. 112]

23. (2)

Mn^{2+} and Fe^{3+} have same electronic configuration i.e. $[\text{Ar}]3\text{d}^5$.

[New NCERT Class 12th Page No. 90]

24. (3)

Ce has maximum composition in misch metal.

[New NCERT Class 12th Page No. 113]

SECTION-(III) BOTANY

25. (3)
VNTR (variable number of tandem repeats) belong to a class of repetitive DNA dispersed in the genome.

[New NCERT class 12th Page No. 106]

26. (1)
Bioactive molecule, cyclosporin A, that is used as an immunosuppressive agent in organ transplant patients, is produced by the fungus, *Trichoderma polysporum*.

[New NCERT class 12th Page No. 153]

27. (2)
With respect to the coding strand, the promoter is present upstream (5' end) of the structural gene in the transcriptional unit.
With respect to the coding strand, the terminator is present downstream (3' end) of the structural gene in the transcriptional unit.

[New NCERT class 12th Page No. 92]

28. (3)
A biological control being developed for use in the treatment of plant disease is the fungus *Trichoderma* species. They are free living fungi that are very common in the root ecosystems. They are effective biocontrol agents of several plant pathogens.

[New NCERT Class 12th Page No. 157]

29. (1)
 - The coding sequence or expressed sequences are defined as exons.
 - Intervening sequences do not appear in processed RNA.

[New NCERT Class 12th Page No. 91, 92, 93]

30. (4)
Broad palm with single palm crease is visible in a person suffering from Down's syndrome.

[New NCERT Class 12th Page No. 76]

31. (3)
 - In Griffith's experiment, a mixture of heat-killed virulent bacteria S and live non-virulent bacteria R, lead to the death of mice.
 - 'Transforming principle' got transferred from heat killed S strain to live R strain and made it virulent.

[New NCERT Class 12th Page No. 84]

32. (2)
Primary treatment involves the physical removal of particles- large and small – from the sewage through filtration and sedimentation. These are removed in stages, initially, floating debris removed by sequential filtration. Then the grit (soil and small pebbles) is removed by sedimentation.

[New NCERT Class 12th Page No. 154]

33. (1)
Sickle cell anemia is caused by the substitution of Glutamic acid (Glu) by Valine (Val) at the sixth position of the beta globin chain of the haemoglobin molecule. The mutant haemoglobin molecule undergoes polymerisation under low oxygen tension causing the change in the shape of the RBC from biconcave disc to elongated sickle like structure.

[New NCERT Class 12th Page No. 74, 75]

34. (2)

<i>Aspergillus niger</i>	Citric Acid
<i>Acetobacter aceti</i>	Acetic Acid
<i>Clostridium butylicum</i>	Butyric Acid
<i>Lactobacillus</i>	Lactic Acid

[New NCERT Class 12th Page No. 153]

35. (1)
The BOD test measures the rate of uptake of oxygen by microorganisms in a sample of water and thus, indirectly, BOD is a measure of the organic matter present in the water.

[New NCERT Class 12th Page No. 154]

36. (3)

XX-XO method of sex Determination	Grasshopper
XX-XY method of sex Determination	Female homogametic
Karyotype-47	Klinefelter's Syndrome
ZW-ZZ method of sex Determination	Female heterogametic

[New NCERT Class 12th Page No. 71, 76]

SECTION-(IV) ZOOLOGY

37. (2)

Genital herpes is a sexually transmitted infection (STI) which is not completely curable.

[New NCERT Class 12th Page No. 47]

38. (2)



- It is poppy plant *Papaver somniferum*.
- Heroin commonly called smack is chemically diacetylmorphine which is a white, odourless, bitter crystalline compound. This is obtained by acetylation of morphine.
- Generally taken by snorting and injection, heroin is a depressant and slows down body functions.

[New NCERT Class 12th Page No. 142]

39. (1)

- In amniocentesis some of the amniotic fluid of the developing foetus is taken to analyse the fetal cells and dissolved substances.
- Statutory ban on amniocentesis for sex-determination to legally check increasing menace of female foeticides, massive child immunisation, etc., are some programmes that merit mention in this connection.

[New NCERT Class 12th Page No. 62]

40. (2)

GIFT (Gamete Intrafallopian Transfer) involves the transfer of gametes (sperm and eggs), not embryos, into the fallopian tube.

[New NCERT Class 12th Page No. 59]

41. (4)

Syphilis, Hepatitis-B, HIV, Genital Warts, and Chlamydia all these are sexually transmitted diseases.

[New NCERT Class 12th Page No. 151]

42. (4)

T-lymphocytes are responsible for rejecting kidney grafts, while B-lymphocytes produce antibodies against smallpox.

[New NCERT Class 12th Page No. 135]

43. (1)

- IUDs is one of the most widely accepted methods of contraception for males in India.
- IUD (Intrauterine Device) is used for female contraception.

[New NCERT Class 12th Page No. 144]

44. (4)

MRI uses strong magnetic fields and non-ionising radiations to accurately detect pathological and physiological changes in the living tissue.

[New NCERT Class 12th Page No. 141]

45. (4)

Lymphocytes interact with antigens and proliferate to become effector cells in the spleen, lymph nodes, Peyer's patches, and tonsils.

[New NCERT Class 12th Page No. 138]

46. (1)

- Physical barriers: Skin on our body is the main barrier which prevents entry of the micro-organisms. Mucus coating of the epithelium lining the respiratory, gastrointestinal and urogenital tracts also help in trapping microbes entering our body.
- Physiological barriers: Acid (HCl) in the stomach, saliva in the mouth, tears from eyes—all prevent microbial growth.
- Cytokine barriers: Virus-infected cells secrete proteins called interferons which protect non-infected cells from further viral infection.

[New NCERT Class 12th Page No. 134]

47. (2)

Coitus interruptus	Withdrawing penis before ejaculation
Cu7	Reduces the motility and fertilising capacity of sperms
Vaults	Prevent physical meeting of sperm and ovum
Vasectomy	Block gamete transport

[New NCERT Class 12th Page No. 44]

48. (2)

- *Wuchereria bancrofti* causes filariasis, not typhoid.
- *Salmonella typhi* is a pathogenic bacterium which causes typhoid fever in human beings.
- These pathogens generally enter the small intestine through food and water contaminated with them and migrate to other organs through blood.

[New NCERT Class 12th Page No. 130]



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